



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VI Semester
End Semester Examination - April / May -2025
AI for Medical Image Analysis [CSE-4011]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the Sequential K-Means algorithm to cluster the following 2D dataset with 6 points:

A)

P1	(1,1)
P2	(2,1)
P3	(4,3)
P4	(5,4)
P5	(8,8)
P6	(9,10)

(5)

Find two (2) clusters by using P1 and P4 as initial cluster centers.

- B) Analyse how domain knowledge is applied differently in hierarchical segmentation versus foreground segmentation. (3)
- C) Explain the difference between push and pull modes of image transfer and how each affects network efficiency and clinical workflow. (2)
- 2) Describe the main steps in Mahapatra et al.'s active learning process for image segmentation, including how the initial model is trained, how uncertainty and diversity guide sample selection, and how a conditional GAN is used to enhance the labelled dataset before retraining. (4)
- A)
- B) Analyse the importance of annotated imaging datasets in building effective machine learning models for radiology, and explore the factors that influence their reliability and availability. (3)
- C) Analyse the role of intensity comparisons in the BRIEF descriptor and explain how they contribute to efficient keypoint matching. (3)
- 3) How can domain knowledge about boundary appearance be applied to improve the segmentation of a foreground object with unclear boundaries in an ultrasound image using methods such as active contours or level sets? (5)
- A)
- B) Apply the concept of temporal continuity to describe how you would segment a 4D (3D+ Time) MRI heart sequence consistently across multiple time steps. (5)
- 4) (5)

- Analyse how the use of both spatial and contextual encoding paths in the Dual Encoding U-Net architecture enables the model to balance local detail preservation with global context understanding for effective retinal vessel segmentation.
- A)
- B) Analyse how the initial low-resolution classification step in the coarse-to-fine ML-assisted tumour detection pipeline contributes to the overall efficiency of the framework. Why is it critical to the system's performance? (5)
- 5) Analyse the architectural differences between TensorFlow and Caffe and discuss how these differences impact their use in designing deep learning models for medical image segmentation. (5)
- A)
- B) Explain how organizing radiology data in a consistent and standardized format can help in the development of machine learning applications. (3)
- C) Explain why a multivariate normal distribution is commonly used for modelling feature vectors. (2)

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